

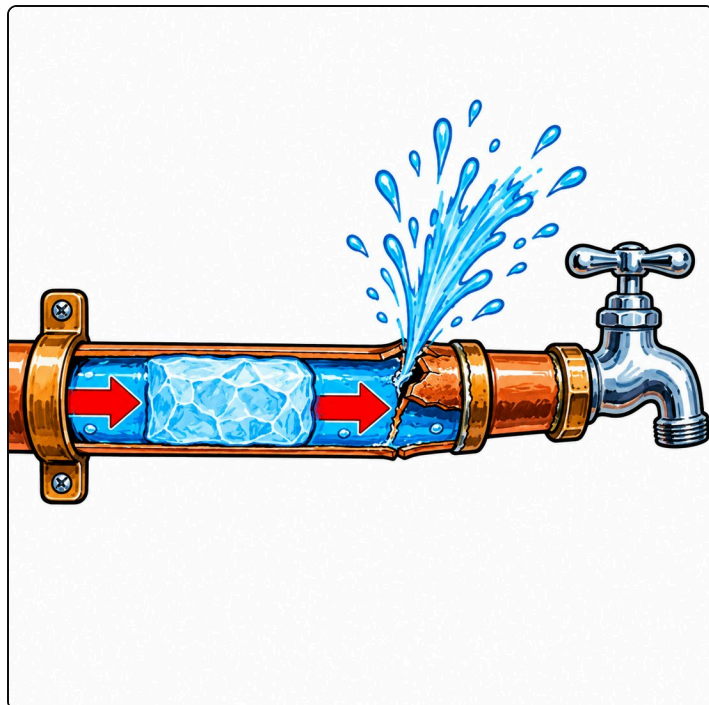
THE \$2 FIX THAT STOPS A \$10,000 BURST PIPE

A frozen pipe almost never bursts where the ice is. That is why a dripping tap and a couple dollars of foam can stop a flood.

THE THREE PART FIX

- ☐ **Foam sleeve**, under 2 dollars a stick, on every pipe that runs through cold space.
- ☐ **A thin, steady stream** from the right faucet on cold nights.
- ☐ **Cabinet doors under your sinks, left open**, so room heat reaches the pipes.

The drip is the part that actually saves the pipe, and most people drip the wrong tap. Once you see how a pipe really bursts, you will know which one.



The ice forms a plug. The split happens downstream, at a joint or thin spot, often where there is no ice at all.

HOW A PIPE REALLY BURSTS

Ice starts as a skin on the inside of the pipe wall, thickens inward, and finally closes the channel into a solid plug. Small pipes freeze first, because they lose their heat fast.

Water grows about 9 percent when it freezes. On the street side of the plug, water can move back toward the main, so no harm. On the other side, a trapped column of water sits against a closed faucet.

40 to 80	Pounds per square inch of normal house pressure.
thousands	Pounds per square inch it takes to burst copper in lab tests.
tens of thousands	Pounds per square inch ice can push. The copper gives up long before that.

Water barely compresses, so the growing plug works like a piston. The pipe splits downstream at the weakest point, often where there is no ice. The plug seals the split while frozen. Then it thaws and full pressure pours through the crack.

1 in 60	Insured homes files a water damage or freezing claim every year, says the Insurance Information Institute.
\$15,400	The average such claim paid, 2019 to 2023.

WHY THE DRIP WORKS

Open the faucet at the end of that trapped column and the water is not trapped anymore. When the plug grows, it pushes water out the tap instead of building pressure.

The drip helps a second way. A moving stream keeps swapping chilled water for fresh water from the main, which runs underground and stays above freezing. Freezing pulls a huge amount of heat out of water, the same as cooling it another 144 degrees, which is why a pipe takes hours to freeze.

Aim for a thin, steady stream, about the width of a pencil lead

One lonely drip relieves pressure but barely moves water. A thin stream overnight costs well under a dollar. A burst pipe costs thousands.

WHICH FAUCET

- ☐ Every tap fed by a pipe you know runs through cold space: a crawlspace, an attic, a garage, or an outside wall.
- ☐ Water only moves toward the tap that is open, so every exposed branch needs its own open tap.
- ☐ If you do not know where your pipes run, open the faucet farthest from where the water enters the house.
- ☐ The Red Cross says to drip the cold side. If a hot line runs through the same cold space, crack that one too.
- ☐ Open the cabinet doors under sinks on outside walls, and keep the thermostat at the same setting day and night.
- ☐ Pipes inside a finished wall usually cannot be sleeved. The open doors and the drip do the work there.

PEX AND SMART VALVES

Pex, the flexible plastic tubing, stretches and survives freezes that split copper. The brass fittings do not stretch, so the same drip rule applies.

A **smart shutoff valve** like a Moen Flo runs about 500 to 700 dollars plus a plumber. It is a good backstop, but it cannot stop ice from forming, and it can read your overnight drip as a leak and shut the main. If you own one, set it to allow the drip on freezing nights.

THE STORM THAT PROVED IT

Winter storm Uri, February 2021, Texas: about half a million insurance claims, about 85 percent of them home property claims, many from burst pipes. Insured losses reached about ten billion dollars. One Waco plumber went from about 40 calls a day to between 120 and 200, and parts took 3 to 10 weeks. A woman near Waco lived nearly a year without running water.

THE FOAM SLEEVE

- 1 Buy the size stamped for your pipe. If unsure, measure across the outside of the pipe and take that number to the store.
- 2 Push the slit sleeve over the pipe and tape the seam. A loose sleeve leaves an air gap, and moving air carries heat away.
- 3 Cut short pieces for the elbows and valves. Butt the ends tight and tape those joints too. Every bare inch leaks heat.

\$1.50

A six foot stick for half inch pipe at the big box stores.

10 to 20

Sticks for most crawlspaces.

\$15 to \$30

Foam and tape for the whole house.



The sleeve slows the heat leaving the water. It adds none, so it goes together with the drip.

WHEN THE DANGER STARTS

University of Illinois field tests found uninsulated attic pipes start freezing when the outside air falls to about 20 degrees. Southern plumbers said burst calls start in the teens. **So 20 degrees is your alarm. When the forecast crosses it, the drip goes on.** Wind makes it worse, so close crawlspace vents for the snap and foam any gaps near the pipes.

THE OUTDOOR FAUCET AND HEAT CABLE

- ☐ Disconnect the hose, even on a frost free spigot. A connected hose traps water inside and is the most common way they fail.
- ☐ Close the inside valve that feeds the line, then open the outside spigot so it drains.
- ☐ For the one pipe that freezes every year, use heat cable. A 30 foot kit runs about 45 to 50 dollars, draws 7 watts a foot, and has a built in thermostat. It goes on the pipe first.
- ☐ Plug heat cable into a ground fault outlet, the kind with test and reset buttons.
- ☐ Never put foam sleeve over heat cable. Use the maker's fiberglass wrap. The cable must never cross over itself.
- ☐ If last year's cable is cracked or brittle, throw it out. Old heat cable starts fires.

IF IT FREEZES ANYWAY

- ☐ If a tap gives nothing, or a trickle, you probably have a plug. Leave that faucet open so the melt has somewhere to go.
- ☐ Find the frozen section, usually along an outside wall or where the line enters the house.
- ☐ Warm it with a hair dryer, a heating pad, or hot towels. Start near the faucet and work back toward the plug. Stay with it the whole time.
- ☐ **Never use a torch or any open flame.** That is how a frozen pipe turns into a house fire.
- ☐ Check the other taps too. If you cannot find or reach the plug, call a plumber before it thaws on its own.

WATCH THE THAW. DO THIS TONIGHT.

- ☐ **Tonight:** find your main water shutoff and make sure it turns. When a pipe splits, how fast that valve closes decides between a wet floor and a gutted kitchen.
- ☐ When the temperature climbs back above freezing, walk the house. Look at ceilings and under sinks. Listen for hissing in the walls.
- ☐ Close every tap and check the water meter. If it keeps turning, close the main right away.

LEAVING TOWN, OR THE HEAT DIES

- ☐ Keep the house at 55 degrees or warmer, or drain it.
- ☐ To drain: turn off the water heater first, gas or electric. Shut the main. Open the highest faucet to let air in, then the lowest to drain the lines.
- ☐ Two open faucets will not empty every low spot. For a long trip, have a plumber winterize the house.
- ☐ The same drain move works when the power dies in a cold snap and the furnace goes quiet for more than a few hours.



A pencil lead stream, cabinet doors open. Cheap insurance against a flood you find on the warm afternoon after the freeze.

Watch the full video: youtube.com/@SketchySurvival101

More free sheets: sketchysurvival101.com/free

Subscribe so the next cold snap does not catch you flat footed.